

Dr Sathish Kumar Marimuthu

Computational cardiovascular biomechanics

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About / Research

Computational cardiovascular biomechanics

I study the mechanics of cardiovascular tissues and devices, with a focus on patient-specific aortic modelling, stent-graft deployment, dissection propagation, and vascular growth and remodelling. My work connects nonlinear continuum, contact and fracture mechanics with clinical imaging and medical-device design.

My current appointments span the University of Glasgow, Aarhus University Hospital and Politecnico di Torino, alongside collaboration with TerumoAortic. I also contribute to research on statistical emulation of nonlinear cardiovascular systems.

Current positions

2025–2028

Research Associate

EPSRC Centre for Future PCI Planning, University of Glasgow & TerumoAortic Ltd, UK

2025–2028

Visiting Researcher

Department of Clinical Medicine — Department of Cardiological Medicine, Aarhus University Hospital, Denmark

2025–2028

Visiting Researcher

PolitoBIOMed Lab, Department of Mechanical and Aerospace Engineering (DIMEAS),
Politecnico di Torino, Italy

Research themes

01 / MECHANOBIOLOGY

Aortic mechanics & adaptation

Anisotropic fibre-reinforced tissue models, dissection propagation, and stent-induced growth and remodelling informed by follow-up CT imaging.

02 / COMPUTATIONAL MECHANICS

Patient-specific intervention modelling

Finite-element analysis of stent designs and virtual stent-graft deployment to investigate device–aorta interactions and post-surgical complications.

03 / SURROGATES & INFERENCE

Cardiovascular emulation

Research contributions on deep Gaussian processes and multivariate Bayesian spline-based emulators for nonlinear cardiovascular systems; related training in material parameter identification.

Clinical & industrial translation

Medical-device collaboration and clinical imaging connect aortic modelling with intervention research; current work is based within the EPSRC Centre for Future PCI Planning.

Publications

2026 **Deep Gaussian processes and multivariate Bayesian spline-based methods for emulating nonlinear cardiovascular systems**

Macaulay C, Husmeier D, Davies V, Marimuthu SK, Colebank M.

Journal of Machine Intelligence and Data Science, 7:09–21.

[DOI: 10.11159/jmids.2026.002](https://doi.org/10.11159/jmids.2026.002)

2025 **A three-dimensional computational study of critical pressures of dissection propagation in the aorta**

Marimuthu SK, Luraghi G, Maclean C, Brodie R, Migliavacca F, McGinty S, et al.

Biomechanics and Modeling in Mechanobiology, 24:1767–1780.

[DOI: 10.1007/s10237-025-01991-2](https://doi.org/10.1007/s10237-025-01991-2)

2021 **Virtual reality-guided left ventricular assist device implantation in pediatric patient: Valuable presurgical tool**

Ramaswamy RK, Marimuthu SK, Ramarathnam KK, Vijayasekharan S, Rao KGS, Balakrishnan KR.

Annals of Pediatric Cardiology, 14(3):388–392.

[DOI: 10.4103/apc.apc_81_21](https://doi.org/10.4103/apc.apc_81_21)

Selected projects

CURRENT RESEARCH

Stent-induced aortic remodelling

Studying vascular growth and remodelling using follow-up CT data, including homogenized constrained mixture theory presented at VPH 2026.

CURRENT RESEARCH

Patient-specific stent-graft mechanics

Virtual deployment in anisotropic aortic models to study stresses, device designs and the risk of post-surgery complications.

SELECTED WORK

Aortic dissection propagation

Three-dimensional computational analysis of critical pressures for dissection propagation, developed through an industrial PhD and published in 2025.

SELECTED WORK · 2026

Cardiovascular system emulators

Co-authored research comparing deep Gaussian processes and Bayesian spline-based methods for emulating nonlinear cardiovascular systems.

SELECTED WORK · IIT MADRAS

Virtual reality for LVAD implantation

Developed a C# / Unity planning and validation tool for left ventricular assist device implantation, with clinical collaboration at MGM Hospitals, Chennai.

Brain shift during craniotomy

Investigated the influence of cerebrospinal-fluid drainage on brain shift and neurosurgical navigation using fluid–structure interaction and image-based tools.

Funding & awards

- **£50,000 · Principal Investigator**
Glasgow Knowledge Exchange Fund (GKEF) and Terumo Aortic, 2026–2027.
- **£71,939 · Co-Investigator**
EPSRC Impact Acceleration Accounts (IAA) & Terumo Aortic grant, 2025–2026.
- **Project Supervisor**
EPSRC Vacation Internship: paid undergraduate summer internship, July–September 2026.
- **Doctoral Researcher**
EPSRC SofTMech and Terumo Aortic PhD studentship, 2020–2024. Grant EP/S030875/1.
- Half-Time Research Assistant Fellowship (HTRA), IIT Madras, 2017–2020.
- Third Prize, Poster Presentation, Soft Tissue Modelling Workshop, 8 June 2023 — design of patient-specific stent-grafts to treat aortic diseases.
- DST-INSPIRE Internship for Young Scientists, Department of Science and Technology, Government of India; Bharathiar University, Coimbatore, 3–8 January 2011.

Presentations

Conference presentations

2026

Modelling stent-induced remodelling in aneurysmal aortas using homogenized constrained mixture theory

Marimuthu SK, Geelkerken R, Brodie R, Maclean C, McGinty S, Hill NA. VPH 2026, Milan, Italy, 1–4 September.

2025

Finite element modelling and validation of stent-induced growth and remodelling in aneurysmal and dissected aortas

Marimuthu SK, McGinty S, Hill NA. BioMedEng25, Glasgow, UK, 4–5 September.

2022

Effect of tapered stent-graft on maximum stress concentration and the reduction in the risk of distal stent-induced new entry (d-SINE) tears

Marimuthu SK, McGinty S, Hill NA. World Congress of Biomechanics, Taipei, Taiwan, 13 July.

2022

Numerical simulation of dissection propagation in three-dimensional patient-specific aortas

Marimuthu SK, McGinty S, Hill NA. World Congress of Biomechanics, Taipei, Taiwan, 12 July.

2022

A methodology to investigate aortic dissection propagation in a three-dimensional aorta

McGinty S, Marimuthu SK, Hill NA. European Solid Mechanics Conference, Galway, Ireland, 5 July.

2021

Understanding the stent-induced stresses in patient-specific aortas modelled with an anisotropic hyperelastic constitutive model

Marimuthu SK, McGinty S, Hill NA. BMC-BAMC, Glasgow, UK, 8 April. Online poster presentation.

2020

Use of Virtual Reality as a Presurgical Planning Tool in LVAD Implantation

Sathishkumar M, Krishnakumar R, Balakrishnan K. 66th ASAIO Annual Meeting, June. ASAIO Journal, 66(Supplement 2), p. 31, CAR11.

Invited talks & seminars

- IAA-INI-HIMR Workshop, University of Strathclyde, Glasgow, 2–3 March 2026 — invited talk.
 - Continuum Mechanics & Industrial Mathematics Group, University of Strathclyde, Glasgow, 26 November 2025 — invited seminar.
 - 2nd Terumo Aortic Academic Collaboration Conference, University of Strathclyde, 18 September 2025 — simulation framework for stent deployment and post-surgical vascular remodelling in aortic dissection and aneurysm.
 - 1st Terumo Aortic Academic Collaboration Conference, University of Strathclyde, 6 June 2023 — optimisation of stent-graft designs to reduce post-surgery complications in aortic dissection treatment.
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Teaching

University of Glasgow

Temporary Lecturer · November–December 2025

James Watt School of Engineering: Engineering Mathematics 1.

Teaching Assistant & Demonstrator · May 2022–August 2023

- Mechanics of Materials 3: solid mechanics for undergraduate, MEng and MS students.
- Mechanics of Materials 4 and Finite Element Methods: ABAQUS labs covering elastic and plastic behaviour, a plate with a hole, and a bearing bracket.
- Design and Manufacture 2; Materials 1 & 2: tensile and impact testing, heat treatment and material behaviour.
- Engineering Skills 1 (Mechanical): miniature trebuchet projects using laser-cut materials.

Indian Institute of Technology Madras

Teaching Assistant · May 2017–August 2020

Department of Engineering Design: Computational Methods in Design and Design of Implantable Devices.

Collaborations

Research collaborations combine computational mechanics, medical-device development and clinical expertise.

- **TerumoAortic Ltd, UK** — industrial PhD, device modelling and knowledge-exchange research.
 - **Aarhus University Hospital, Denmark** — visiting research in patient-specific stent modelling and vascular remodelling.
 - **Politecnico di Torino, Italy** — PolitoBIOMed Lab visiting appointment and research collaboration.
 - **Politecnico di Milano, Italy** — aortic device modelling collaboration.
 - **University of Oxford, UK** — research collaboration listed within the current Research Associate role.
 - **MGM Hospitals, Chennai, India** — cardiothoracic clinical collaboration on LVAD planning.
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Education & training

- **Industrial PhD, Applied Mathematics · April 2024**
University of Glasgow & TerumoAortic Ltd. Thesis: "Optimisation of stent devices to treat dissected aorta." Supervisors: Nicholas A Hill and Sean McGinty; industry supervisors: Robbie Brodie and Craig MacLean. Collaborators: Francesco Migliavacca and Giulia Luraghi.
- **MS, Engineering Design · July 2020**
IIT Madras. Thesis: "Mechanistic modelling and virtual reality in improving the outcome of surgeries." Supervisor: R Krishna Kumar; collaborator: Komarakshi Balakrishnan.
- **BEng, Mechanical Engineering · May 2016**
Anna University, India.

Professional training

- Biomechanics of Soft Tissues: From the Heart to the Cardiovascular System to the Brain — Technical University of Graz, Austria, 11–15 September 2023. Continuum mechanics, nonlinear elasticity, image-based modelling, vascular adaptation and parameter identification.

- Computational Tissue Biomechanics: From in-vitro experiment to damage and failure analysis — KTH Royal Institute of Technology, Sweden, 5–9 September 2023. Included biaxial tensile testing of porcine aorta for material parameter estimation and curve fitting.
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Skills

Mechanics & simulation

Nonlinear continuum mechanics · contact mechanics · fracture mechanics · fluid–structure interaction · ABAQUS / Explicit · ANSYS

Programming

Python · FORTRAN for user-defined materials · MATLAB · C# / Unity Engine

Imaging & geometry

3D Slicer · ITK-SNAP · MeshMixer · BrainSuite · Ansys SpaceClaim · Autodesk Fusion 360 · SolidWorks

Research practice

Patient-specific modelling · material parameter estimation · biaxial tissue testing · HPC server maintenance · English (professional) · Tamil (native)

Professional service

Student & Early Career Researcher representative roles at the University of Glasgow: Centre for Mathematics Applied to the Life Sciences; Applied Mathematics, School of Mathematics and Statistics; and the Graduate School, College of Science and Engineering.

Contact

Glasgow, UK · mail2sathish.skm@gmail.com

Earlier experience

Research Assistant and Lab In-charge · 2017–2020

Indian Institute of Technology Madras. Fluid–structure interaction and virtual reality for neuro and cardiothoracic surgeries. Developed LVAD planning tools, studied brain shift during craniotomy, and maintained HPC servers for the Artificial Intelligence in Medical Imaging group at the Trivitron Innovation Centre for Medical Technology.
